Launch Assessment - Detailed Analytical Report

New Product - Biscuits & Snacks - launch 2025-03-03

DEMONSTRATION: models and market data are SIMULATED. Figures illustrate the workflow and are not real-world observations.

Recommendation: Run a limited regional pilot. P(hurdle) = 83% looks favourable, but model reliability is Low (see risks), so validate demand on a small scale first.

Decision-support estimates only - not guarantees of commercial success. Values are labelled as Observed (historical data), Model estimate, or Assumption (user-entered).

1. Proposed product

Field	Value
Name	New Product
Category	Biscuits & Snacks
Positioning	mass market
Pack	100 g (Pouch)
Shelf price (incl. tax)	INR 35.00
Target regions	North, South
Outlets	{"Hypermarket": 20, "Supermarket": 80, "Convenience": 150}
Launch date	2025-03-03

2. Headline results

Metric	Value
Commercial hurdle attainment probability [Model estimate]	83% (target profit 0 after launch costs within 26w)
Expected units 1 / 3 / 6 / 12 months [Model estimate]	24,578 / 156,609 / 415,832 / 937,199
Expected 6-month manufacturer net revenue [Model estimate]	INR 10,135,914
Expected 6-month contribution after launch costs [Model estimate]	INR 2,117,495
Unit contribution / margin [Calculated from assumptions]	10.39 / 42.6%
Break-even units (regular economics) [Calculated]	211,793
Model reliability [Validation-based]	Low (score 40)

3. Hurdle definition

A launch 'meets the commercial hurdle' when cumulative contribution over the evaluation window (default 26 weeks), AFTER deducting all fixed launch spend, is >= the user's target profit (default 0 = break-even). Units lost to stock-outs are not counted. Error calibration only uses products with enough observed weeks.

4. Unit economics (manufacturer view) [Calculated from user assumptions]

Per unit	Regular	Introductory period
consumer price	35.00	35.00
price ex tax	31.25	31.25
trade margin amt	6.87	6.87
discount cost	0.00	0.00
net revenue	24.37	24.37
wastage	0.49	0.49
variable cost	13.99	13.99
contribution	10.39	10.39

5. Demand and financial projection [Model estimate; P10-P90 from out-of-fold launch errors]

Horizon	Units (mean)	Units P10-P90	Net revenue	Contribution	After launch costs
1 month	24,578	9,434 - 45,847	599,078	255,110	-1,944,890
3 months	156,609	62,762 - 291,868	3,817,336	1,625,583	-574,417
6 months	415,832	182,616 - 695,000	10,135,914	4,317,495	2,117,495
12 months	937,199	450,197 - 1,345,000	22,844,223	9,732,092	7,532,092

6. Scenarios

Scenario	Quantile	Demand	Cost	Avail.	Units 6m	Contribution 6m	After launch costs	Hurdle
Conservative	P10	x0.48	x1.05	x0.90	178,132	1,730,110	-469,890	No
Base	P50	x0.82	x1.00	x1.00	338,381	3,514,928	1,314,928	Yes
Optimistic	P90	x1.75	x0.97	x1.00	695,000	7,500,787	5,300,787	Yes

7. Break-even, inventory and cash

Item	Value
Break-even units (regular economics)	211,793
Break-even revenue (manufacturer net)	5,162,455
Break-even week (median path)	19
P(break-even within 12 months)	99%
Planned vs recommended (P90) opening inventory	120,000 vs 29,769
P(stock-out) / expected lost units	17% / 41,985
Peak funding need (median path)	INR 5,659,494 (week 16)
Budget utilisation	79% of total launch budget

8. What-if analysis (scenario estimates, not causal effects)

Scenario	P(hurdle)	Expected after launch costs	Units 6m
Current plan	83%	2,117,495	415,832
Price -5%	81%	1,859,427	441,218
Price +10%	84%	2,432,791	362,364
Marketing budget +20%	74%	1,734,363	415,023
Distribution coverage x2	97%	3,876,466	584,842
Launch delayed 4 weeks	84%	2,124,655	416,015
Competitor prices -10%	81%	2,005,592	404,551
Demand 15% below forecast	74%	1,601,371	365,652
Manufacturing cost +10%	76%	1,688,243	413,814

Sensitivity (+/-10% one-at-a-time)

Driver	-10% change in result	+10% change in result
Demand level	-351,493	351,493
Distribution (outlets)	-343,623	341,655
Manufacturing cost	338,381	-338,381
Shelf price	-235,508	119,130
Marketing budget	180,059	-182,036

10. Risks and reliability

[medium] On held-out products the model did not beat the simple category baseline; treat point forecasts cautiously.

[medium] Out-of-sample launch totals differ from forecasts by a typical factor of exp(0.54) = x1.71.

[low] Uncertainty was calibrated on 26-week totals; 12-month ranges are likely understated.

[high] DEMO MODE: models were trained on SIMULATED data; outputs illustrate the workflow only.

11. Recommendations

Area	Action	Why	Evidence	Trade-off	Source / confidence
Launch timing	Consider delaying launch by about 26 weeks	The demand model's seasonal pattern favours a later start.	Median-path contribution after launch costs: INR 1,314,928 now vs INR 2,032,685 with a 26-week delay.	Delay carries competitive and cash-flow costs not modelled here.	model-derived / Low
Inventory planning	Reduce the opening stock build or phase production	65% of simulated outcomes end the year with more than 4 weeks of stock.	Expected end inventory 407,801 units.	Higher stock-out risk if demand surprises upward.	model-derived / Medium
Risk mitigation	Pilot in one region / channel before national rollout	Forecast uncertainty and/or data-support issues are material.	DEMO MODE: models were trained on SIMULATED data; outputs illustrate the workflow only.	Slower rollout.	rule-based / High

12. Data sources, freshness and model evaluation

Dataset: Synthetic FMCG demo dataset (SIMULATED) (mode: demo_synthetic), 2023-01-02 to 2024-12-23.
Launch-outcome labels: NO - a launch-success classifier is not trained; hurdle probability is a scenario estimate.

Cold-start model (held-out products)	WAPE	MAE	Product MdAPE 26w
Baseline	0.500	28.0	0.28
Poisson GLM	0.492	27.5	0.27
Random Forest	0.579	32.3	0.42
HistGradientBoosting (Poisson)	0.536	29.9	0.32
LightGBM (Poisson) [selected]	0.532	29.7	0.32

Uncertainty: 60 out-of-fold launches; SD of log(actual/forecast) of 26-week totals = 0.54; median ratio x0.82.

13. Assumptions requiring validation

- Marketing response: max uplift 20%, half-saturation 2,000/outlet, post-campaign retention 50% (assumption; no campaign data).

- Price response: within-product elasticity -1.82 (observational).

- Cross-price elasticity 0.3 (assumption); competitor price change 0.0%.

- Tax 12%, trade margin 22%, wastage 2%, receivable days 30.

- Unit cost realisation uncertainty SD 3% (assumption).

- Scenario cost (+5%/0/-3%) and availability (0.9/1/1) multipliers are manual.

- Per-outlet demand is taken from sampled historical stores; planned outlets are assumed comparable.